Defense-in-Depth Under Military Action

DBA-MA-DiD

A Correlated-Barrier Adequacy Methodology for Deliberate Action Against Critical Infrastructure

Shared Machinery of the DBA-MA Extension · Inherited by Every Sector

Version 0.4 — WORKING DRAFT

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Sector-neutral method — inherited by the sector frameworks, facility classes, use cases and reference events of every sector, and applied to the nuclear sector by DBA-NUC-DiD (Defense-in-Depth Under Military Action: Nuclear Sector Application).

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Note on This Version

Versions 0.1 (July 2026) and 0.2 (22 September 2026) of this instrument carried the method as written for nuclear facilities, under the subtitle *A Correlated-Barrier Adequacy Methodology for Deliberate Action Against Nuclear Facilities*. The referent is unchanged. DBA-MA-DiD names the same method it has always named: the barrier-independence ledger, the adversary-selected cutset analysis and the independent protection of the governing cutsets, ending in the adequacy statement. What changes at version 0.3 is the method's scope, which is now every sector of the extension, and its place, which is now the shared machinery of the DBA-MA parent beside DBA-MA-AF (*MA Applicability Framework*) and DBA-MA-CDB (*Military Action Communications Design Basis*).

The nuclear application moved at this version to DBA-NUC-DiD (*Defense-in-Depth Under Military Action: Nuclear Sector Application*), which the nuclear series now cites for it. Section numbers in this instrument do not move. Every subsection whose text left keeps its number and carries a one-line note naming where the text went, and new sector-neutral material takes new numbers at the end of the section it belongs to. A citation to this instrument by designation and section therefore resolves to the same matter at every version, including from copies that can no longer be revised.

The moved sections are these. Section 4.3, expressibility as an SSR-2/1 defense-in-depth extension, is now DBA-NUC-DiD §3. Section 4.4, the relationship to the Conditionality Check and the three-tier adversarial design basis, is now DBA-NUC-DiD §4. The worked illustration on the multi-module SMR campus that formed Section 5 is now DBA-NUC-DiD Section 5, with §5.1 to §5.3 at DBA-NUC-DiD §5.1 to §5.3 and the validation against Zaporizhzhia of §5.4 at DBA-NUC-DiD §5.4. Sections 6.1 to 6.3, the application across the nuclear series, are now DBA-NUC-DiD §6.1 to §6.3. Open items §7.4 and §7.5 are now DBA-NUC-DiD §7.2 and §7.3, and the nuclear sequencing record formerly in §7.2 is DBA-NUC-DiD §7.1. The new sector-neutral material is §3.6 on what robust, diverse and independent means, §4.5 on a sector's adversarial design basis and consequence analysis, §5.5 on the worked examples in the sectors, §6.4 on application across the sectors, and §7.8 on whether the four dependency classes suffice in every sector.

1. Purpose, Scope, and Position in the Series

1.1 Purpose

This instrument develops the methodological concept that the DBA-MA extension identifies as a core technical challenge in every sector but that the parent framework states rather than resolves: defense-in-depth under military stress. Traditional defense-in-depth credits a succession of independent barriers, each reducing risk so long as the barriers fail independently of one another. The idea came to critical infrastructure from nuclear safety, and every sector that protects a function with layers now relies on it in some form, whether it calls the layers barriers, redundancy, backup or diversity. Deliberate military action can defeat several barriers at once, or in a chosen sequence, through mechanisms that the correlated-failure analysis of traditional design-basis practice was not built to represent. This document states precisely why the traditional treatment falls short when the initiating condition is intentional, and it supplies a method — the adversary-selected cutset analysis — that restores a rigorous, auditable adequacy statement for defense-in-depth under deliberate action. The method is deterministic and bounding: it asks whether a feasible adversary action set within the facility's design-basis threat tier collapses a credited barrier combination, not what the probability of such an action might be.

1.2 What This Instrument Is, and Is Not

This is a cross-cutting methodological instrument and a member of the shared machinery of the DBA-MA extension. It develops one methodology — correlated-barrier adequacy under deliberate action — and is referenced by the documents that credit defense-in-depth rather than re-derived in each of them. It does not define the adversarial threat tiers, which each sector's facility classes define for themselves; it does not state any sector's consequence analysis, which belongs to that sector's framework and facility classes; it does not own the artificial-intelligence governance apparatus, which is governed by the Command Broker specification family and by each sector's annex to it; and it does not carry the application of the method to any one sector, which the sector writes. The nuclear application is DBA-NUC-DiD. This instrument supplies the barrier-correlation method those documents apply whenever they assert that defense-in-depth is adequate or trace a consequence chain through a set of credited barriers.

It is equally important to state what the method is not. It is not a probabilistic risk assessment, and it does not compete with or replace probabilistic risk assessment. It is not a replacement for common-cause failure analysis. It is not a physical-protection plan and contains no targeting information. It sits at the safety-security interface: it imports the deterministic adversary-capability reasoning long established in the design-basis-threat world into the defense-in-depth adequacy question that safety analysis has historically treated as a matter of random and systematic failure. The contribution is to make that import disciplined, auditable, and expressible within existing safety-assessment guidance.

1.3 Position in the Hierarchy

This instrument is positioned explicitly within the DBA-MA architecture, as the extension requires of every new document.

Type. Shared machinery of the DBA-MA parent. It is none of the five document types the series recognizes on its own account. It is a method, inherited by instruments of each of those types, in the way the applicability gate and the communications design basis are.

Parent. The DBA-MA extension, through its series root, A1 (*An Introduction to the Framework*). This instrument inherits the four characteristics of military action — intentionality, persistence, precision and denial — and the requirement that the consequence chain be traced end to end, and it states the method in terms that each sector's existing defense-in-depth or redundancy practice can take up as an extension rather than a replacement.

Sector applications. DBA-NUC-DiD applies the method to the nuclear sector. It states the method's expressibility under the IAEA's defense-in-depth requirements, its relationship to the nuclear series' source-term and adversarial design basis instruments, its worked illustration on the multi-module SMR campus and the validation of that illustration against Zaporizhzhia, and its application across the nuclear facility classes, use cases and national children. Another sector may write an application of its own on the same terms. None is required, and a sector instrument may cite this method directly.

Referencing instruments. The sector frameworks, facility classes, use cases and reference events of every sector that credit defense in depth or redundancy against deliberate action. The water series credits redundancy only where it is robust, diverse and independent, which is the third move of this method stated in the author's own term (§3.6). The reference-event method allows a reference event to offer defense in depth as a countermeasure where the event shows layers that shared a substrate or failed together, citing this instrument by its moves (§6.4).

Inherited versus locally defined. The following table records the boundary.

Inherited (referenced by owner)	Locally defined (this instrument)
The four characteristics of military action — intentionality, persistence, precision, denial (DBA-MA parent)	The barrier-independence ledger — the four-class dependency schema
The adversarial design basis and threat tier of each facility class (the sector's facility-class instruments)	The adversary-selected cutset analysis — minimal substrate-defeat sets screened by feasibility
Each sector's consequence analysis, and in the nuclear sector the Conditionality Check of DBA-MA-ST, by way of DBA-NUC-DiD	The independent-protection adequacy criterion for the governing cutsets, and what robust, diverse and independent requires
The Command Broker specification family and each sector's annex to it, where a credited barrier is a governance barrier (CB-REG)	The DiD-adequacy statement — the deterministic, bounding form of the result

1.4 Relationship to Governing Documents

This instrument is issued as shared machinery of the DBA-MA extension. It draws the adversary capability for its feasibility screen from the adversarial design basis of the facility class to which it is applied. Where a credited barrier is a governance barrier — a human authorization interposed on a non-deterministic system — it references the Command Broker specification family, registered in CB-REG, and the sector's annex to it, rather than redefining either. Where it is applied to the nuclear sector, it is applied through DBA-NUC-DiD. Version numbers are not pinned in this document; each reference resolves against the owner's authoritative mirror in accordance with the portfolio version rule.

2. The Problem: Why the Traditional Treatment Falls Short Under Deliberate Action

2.1 Traditional Defense-in-Depth and Its Quantitative Expression

Defense-in-depth provides successive, ideally independent, barriers so that no single failure — and no credible combination of failures — defeats the protected function. The concept was formalized in nuclear safety (INSAG-10) and has since been carried, under various names, into the protection practice of every sector that depends on layered barriers. Its quantitative expression is probabilistic risk assessment. Fault-tree analysis reduces a top event, such as the loss of a protected function or an uncontrolled release, to its minimal cut sets — the minimal combinations of basic events whose joint occurrence produces the top event (the fault-tree method of NUREG-0492, in the lineage of the Reactor Safety Study). Correlation among nominally independent components is then handled by common-cause failure modeling (NUREG/CR-5485), through the beta-factor, multiple-Greek-letter, or alpha-factor parameterizations. Each parameter estimates, from operating experience, the fraction of a component's failures that occur in common with its redundant partners. The apparatus is sound and mature for the domain it was built for: random and systematic failures whose correlation structure is stationary and estimable from history.

2.2 Three Ways Deliberate Action Breaks the Assumption

2.2.1 The correlation is selected, not sampled

Common-cause failure parameters are estimated as the probability that redundant components fail together at random. An adversary does not draw from that distribution. It inspects the barrier set and deliberately selects the combination whose joint defeat maximizes consequence for minimum effort. The governing quantity is therefore not the probability that the barriers fail together given random operation, but whether a feasible adversary action set defeating the credited combination exists at all. A common-cause parameter fitted to operating experience has no term for an intelligence that chooses the correlation, and no amount of additional operating

experience will supply one, because the correlation is not a property of the equipment population — it is a property of the adversary's objective.

2.2.2 Independence is defeated at the shared substrate, not at the barrier

A facility's per-unit barriers may be genuinely independent at the unit level — which is exactly what a single-unit risk assessment shows, and exactly why that result misleads when it is carried to the whole facility or to a system of facilities. The barriers rest on shared substrate: the power supply and its interconnection, including the switchyard or substation; the water, fuel or other supply on which the process depends; the shared digital instrumentation-and-control architecture; and the security and response organization. Deliberate action attacks the substrate, converting nominally independent barriers into a correlated set through a dependency path that the barrier-level or unit-level model never traverses. The adequacy question migrates accordingly: it is no longer "are the barriers independent?" but "is the substrate on which their independence rests itself independently protected against deliberate defeat?"

2.2.3 The hidden common cause is the clock

Wherever a barrier holds for a duration rather than indefinitely — a stored reserve, a battery, a tank, a coping time — redundancy in space becomes an argument in time: the barrier holds for so long. On a multi-unit site, or across a set of facilities that draw on the same reserve, every unit's clock starts on the same initiating condition and expires after the same design duration. An adversary who controls the timing of the initiator and, through persistence, whether denial of access outlasts the holding duration, controls a correlated time-to-failure across the whole site. Steady-state common-cause failure modeling, which characterizes the probability of coincident failure, does not represent a synchronized, adversary-controlled countdown. The clock is a common-cause element that is invisible precisely because it is not a component.

2.3 The Consequence for the Adequacy Argument

Under deliberate action, the adequacy of defense-in-depth cannot be certified by demonstrating barrier independence and bounding the random common-cause contribution. It must be certified by demonstrating that no feasible adversary action set within the facility's design-basis threat tier collapses a credited barrier combination without defeating an element that is independently protected. That is a deterministic, bounding statement rather than a probabilistic one, and the remainder of this document is the method that produces it.

3. The Method: Adversary-Selected Cutset Analysis

3.1 Overview

The method has three moves. Each is expressed inside the risk-informed apparatus that the regulator or the owner already uses — dependency mapping, fault-tree cut-set construction, and

the deterministic adversary definition of the design-basis threat — so that the result is auditable by the existing safety-analysis organization rather than through a parallel process. The first move maps credited barriers to the substrate they depend on. The second enumerates the minimal substrate-defeat sets that collapse a credited barrier combination and screens them for feasibility against a defined adversary. The third establishes the adequacy criterion the surviving sets must satisfy.

3.2 Move 1 — The Barrier-Independence Ledger

For each credited barrier, record the substrate on which its safety function depends, sorted into four dependency classes. The physical class covers structures, standoff distance, and siting geometry. The electrical class covers offsite power, the switchyard, buses, and direct-current systems. The digital class covers instrumentation-and-control channels, networks, shared platforms, and historians. The organizational class covers staffing, the security force, the emergency-response organization, procedures, and site access. For each substrate element, record whether it is shared across barriers or across units. The ledger is the physical-protection analog of a common-cause component group, but it is organized by dependency rather than by similarity of component — because deliberate action correlates barriers through shared dependency, not through shared manufacture. The output of Move 1 is a map from each credited barrier to the substrate elements whose defeat removes it.

Where the ledger records a substrate element as shared, the barriers depending on it are dependent. Take the transitive closure of that relation across the credited barrier set: the result partitions the set into **independence classes**, and the count of classes — not the count of barriers — is the redundancy the configuration actually has against an adversary who selects correlations. The closure is conservative by construction, so that a configuration earns an independence claim by declaring the dependencies that would defeat it rather than receiving one by default. An undeclared dependency does not reduce the count; it removes the dependency from view and leaves the configuration scoring as though the independence were real. **There is no analytical remedy for this and the remedy is procedural: the ledger is written, and reviewed by someone other than its author.** A count obtained from a thin declaration is worse than no count, because it is a flattering number carrying a stated derivation.

The count is reported as a bounded range and not as a single figure, because a real declaration is built largely from estimates. Some ledger entries are close to fact — a shared switchyard, heat sink, or substation. Others are judgments — whether two control implementations are independent, whether an update path stops at one group. The closure is taken with the judged entries resolved each way: the upper bound treats every judged independence as holding and the lower bound treats every one as failing, and **the judgments that move the count between the bounds are named with it**, in the form *three independence classes if the two control implementations are independent, one if they are not*. A single figure is the case in which no

judgment moves the result, and it is reported as a single figure only when the review has shown that. The convention for producing the range is open at Section 7.6.

3.3 Move 2 — Adversary-Selected Cutset Analysis

From the ledger, enumerate the minimal sets of substrate elements whose joint defeat collapses the credited barrier combination — the adversary-selected cutsets. This is the fault-tree minimal-cut-set construction of established practice (NUREG-0492), with two disciplined changes. First, the basic events are substrate defeats rather than component failures, so the tree traverses the dependency paths recorded in Move 1 rather than the equipment failure modes. Second, the resulting cutsets are not ranked by probability; they are screened by feasibility against a defined adversary. A cutset is in scope if a single coordinated adversary operating within the facility's design-basis threat tier — the adversarial design basis of the facility class, evaluated at the threat tier assigned to the site — can execute it. This is the deterministic posture that the design-basis threat has long imposed in nuclear physical protection (IAEA Nuclear Security Series No. 10), which this extension carries into every sector: one designs to defeat a defined adversary capability, not to render an attack improbable. The governing cutsets are the minimal, feasible substrate-defeat sets that collapse a credited combination. They are typically few, and they typically concentrate in the shared, above-grade substrate that the ledger exposes.

3.4 Move 3 — Independent Protection of the Governing Cutsets

Defense-in-depth adequacy under deliberate action is not restored by adding redundant barriers. An adversary who selects correlations simply re-selects to include the added barrier, and the redundancy that would reduce random risk adds nothing against an intelligence that chooses its targets. Adequacy is restored instead by ensuring that every governing cutset contains at least one element that is independently protected against defeat at the facility's threat tier. Independent protection takes a different form for each dependency class: standoff hardening for a physical element, such as passive standoff barriers for above-grade structures and equipment; segregation, determinism, and formal verifiability for a digital element, together with the authorization discipline of the Command Broker, as the sector's annex applies it, where a governance barrier is involved; and denial-path closure for an organizational element, such as staffing and access arrangements engineered to survive the duration for which the credited barriers must hold. The load-bearing word is independent: a protecting measure counts only if it does not itself appear in the same cutset — that is, only if it is not defeated by the same action set that defines the cutset. Where an instrument credits redundancy within a governing cutset, it credits it only where the redundancy is robust, diverse and independent, in the sense §3.6 gives those words; redundancy without those properties is the added barrier this section opens by rejecting.

Where a credited barrier is delivered by more than one instance, **the credited consequence is evaluated at the independence class and not at the instance.** A class that fails together also recovers together, through one spare pool and one crew, so the shared substrate extends the

outage as well as widening it — which is the persistence characteristic reaching the adequacy question directly rather than through the coping clock alone. The consequence carried into the adequacy statement is therefore the aggregate across the class under a single correlated defeat, not the per-instance figure. Where the class-level consequence exceeds the criterion, it is disposed of under this Move — by independent protection of at least one element of every governing cutset, or by a declared residual — **and not by adding instances**, for the reason this section opens with.

A governing cutset that contains no independently protected element is not adequate, and its consequence is carried forward as a declared residual rather than assumed away.

3.5 The DiD-Adequacy Statement

The method yields a single, auditable statement of the following form: for the facility class at the stated threat tier, no adversary-selected cutset within the design-basis threat collapses a credited barrier combination without defeating an independently protected element; the cutsets for which this does not hold are enumerated and carried as declared residuals. This is the deterministic, bounding analog of the frequency-based adequacy statement that probabilistic risk assessment produces for the random-failure domain, of which the large-early-release frequency of nuclear practice is the type. It is the form that a facility-class adequacy result or a use-case consequence chain cites when it asserts that defense-in-depth is adequate under military action, and the form a reference event cites when it offers defense in depth as a countermeasure. Because it enumerates its own residuals, it is falsifiable and reviewable: a regulator or reviewer can challenge the ledger, the feasibility screen, or the claim of independent protection for any cutset, and the analysis responds at the level of the specific dependency rather than by appeal to an aggregate probability.

3.6 Robust, Diverse and Independent

The water series credits redundancy only where it is robust, diverse and independent. The phrase is the author’s, and this instrument adopts it as the plain statement of what Move 3 asks of any redundancy it credits. Each word carries its own obligation, and none of them is met by meeting the others.

Robust means that the element holds against the action it is credited against, at the facility’s threat tier: that action does not defeat it. Diverse means that the redundant elements do not share a design, an implementation, a supplier, a toolchain or an update path, so that no single action defeats them together because they are in substance the same thing. This is diversity in the sense nuclear safety has long given it, as the defense against common-cause failure, and it applies in every sector. Section 7.7 holds the open question of when diversity of implementation counts as independent protection. Independent means that the element does not appear in the same governing cutset as the element it is credited to protect, which is the load-bearing test of §3.4: the redundant element is not defeated by the action set that defines the cutset.

Robust is not resilient, and an instrument citing this method does not write one in place of the other. Resilience is the capacity to absorb a loss and recover from it, and it is shown after a barrier has failed. Robustness is the property that the barrier does not fail, and it is shown before. A resilient configuration may recover well from exactly the correlated defeat this method exists to prevent, and recovery does not discharge the adequacy statement of §3.5. The class-level evaluation of §3.4 is where the duration of recovery enters the analysis, as part of the consequence of the defeat and not as protection against it.

4. Relationship to Established Methodology

4.1 An Extension of Common-Cause Failure Analysis, Not a Replacement

Probabilistic risk assessment and its common-cause failure modeling continue to govern the random and systematic failure domain without modification. This method governs the adversary-selected correlation that the common-cause parameters cannot represent, for the reason given in Section 2.2.1: the correlation is a property of the adversary's objective, not of the equipment population. The two are complementary and both are required. A facility carries a frequency-based result for the random domain — in nuclear practice the core-damage and large-early-release frequencies — and a DiD-adequacy statement for the adversarial domain; neither substitutes for the other, and the boundary between them is the boundary between sampled correlation and selected correlation.

4.2 The Deterministic, Bounding Complement — No Attack-Probability Estimation

The method deliberately estimates no attack probability. This is a feature rather than a limitation. Regulators have consistently and rightly distrusted attack-probability numbers, and the design-basis-threat framework has avoided them for decades by specifying an adversary capability to be defeated rather than a likelihood to be tolerated. This method inherits that discipline exactly. Its feasibility screen is a capability test — can a defined adversary execute this cutset? — not a frequency estimate, and its adequacy statement is a defeat test, not a residual-risk tolerance. The result is directly compatible with the physical-protection posture the regulator already applies, and it does not reintroduce, through the safety analysis, the attack-likelihood estimation the security analysis has always refused.

§4.3–§4.4 — Nuclear-specific material, moved at v0.3 to DBA-NUC-DiD §3–§4.

4.5 Relationship to a Sector's Adversarial Design Basis and Consequence Analysis

The method closes a loop that each sector opens when it states its design basis. The sector's framework and facility classes identify the barriers the design basis credits and the consequence

that follows from their defeat, and the facility class defines the adversary capability at each threat tier. The barriers a facility class credits are the inputs to the ledger of Move 1. The four characteristics of military action, applied through the facility class's adversarial design basis, are the feasibility screen of Move 2. The residual a sector analysis is required to state is the output of Move 3.

Where a sector carries a formal check of the kind the nuclear series carries in the Conditionality Check of DBA-MA-ST (*Source Term Under Military Action*), the check states what must be examined and why, and this instrument states how the examination is computed and what adequacy means once it is; DBA-NUC-DiD §4 records that relationship for the nuclear case. Where a sector carries no such check, the method applies directly to the barriers the facility class credits. A class that has not yet defined its adversarial design basis cannot be screened, which is the sequencing dependency of §7.2.

5. Worked Illustrations

§5.1–§5.4 — Nuclear-specific material, the worked illustration on the multi-module SMR campus and its validation against Zaporizhzhia, moved at v0.3 to DBA-NUC-DiD Section 5 at the same numbers.

5.5 Worked Examples in the Sectors

The method's worked illustrations are now carried by the sectors that apply it rather than by this instrument. The nuclear illustration, on the multi-module SMR campus and validated against Zaporizhzhia, is DBA-NUC-DiD Section 5, and it remains the fullest working of all three moves through to an adequacy statement with its residual. Three reference events apply the method to events that occurred, and they are the worked examples the reference-event method points to: DBA-W-RE-1 (*Minnesota Water Campaign*) at its finding F-2, DBA-CHEM-RE-3 (*Graniteville Chlorine Release*) at F-1, and DBA-TRAN-RE-17 (*Chatsworth Collision*) at F-7. *The Asset and the Target*, the energy companion paper, applies the independence-class count of §3.2 and the class-level consequence evaluation of §3.4 in the energy sector. This instrument cites them and does not restate them.

6. Application Across the Series

§6.1–§6.3 — Nuclear-specific material, moved at v0.3 to DBA-NUC-DiD §6.1–§6.3.

6.4 Application Across the Sectors

A sector framework that names defense in depth as a design-basis principle cites this instrument for the method, as the water framework does at its §4.1, and states in its own terms which of its facility classes must produce an adequacy statement. A facility class applies the three moves to

the barriers it credits. It builds the ledger from its own dependencies, screens the cutsets against its own adversarial design basis, and states its adequacy result and its residuals in the form of §3.5. Only the ledger contents and the feasibility screen change from class to class and from sector to sector; the schema, the deterministic posture of the screen and the adequacy criterion do not. A use case applies the method to a specific consequence chain, building the ledger for the barriers on that chain.

A reference event may offer defense in depth as a countermeasure, and it is never required to. Under the amendment to the reference-event method made in *Defense in Depth Is Sector-Neutral, and Reference Events May Offer It* (7 November 2026), it does so where the event shows protective layers that shared a substrate or failed together. The countermeasure is a recommendation, numbered and registered with the instrument's other recommendations. It cites this instrument by its moves, at §3.2 to §3.5, rather than restating them, and where it calls for redundancy it asks for redundancy that is robust, diverse and independent (§3.6). Where an event shows nothing of the kind, the reference event says nothing about defense in depth.

A sector may write an application instrument for the method, as the nuclear sector has in DBA-NUC-DiD, where its regulator expects the method in the regulator's own terms or where its worked illustration is substantial enough to stand on its own. A national child applies the method by way of its sector, supplying the national adversary definition and the national process to which the adequacy statement attaches.

7. Gaps and Open Items

7.1 Ledger Schema Registration

The four-class dependency schema of the barrier-independence ledger should be registered as a formal series template in the portfolio Gaps and Recommendations document, so that facility-class, use-case, and national authors build ledgers with a common structure and a common vocabulary.

Amended at version 0.2. The declared dependency set this item asks for now has a defined output and a defined discipline. Section 3.2 takes the transitive closure of the ledger's shared-substrate relation into independence classes, reports their count as a bounded range with its deciding judgments named, and requires the ledger to be reviewed by someone other than its author. The registration itself — entry of the four-class schema as a formal series template in the portfolio Gaps and Recommendations document — remains to be made. What has closed is the question of what the template is for.

7.2 Feasibility-Screen Dependence on the Adversarial Design Basis

The feasibility screen of Move 2 depends on an adversarial design basis being defined for the facility class in question. Extending the method to a class requires that the class carry its

adversarial design basis first; this is a sequencing dependency, not a methodological gap. Which classes carry one is recorded by each sector, and the nuclear record that stood in this item at version 0.2 is now DBA-NUC-DiD §7.1.

7.3 The Quantification Boundary

The method is deterministic by design. Whether, and how, a bounding likelihood should be attached at the design-basis-threat boundary for risk-informed integration with the frequency-based results of probabilistic risk assessment — in nuclear practice the core-damage and large-early-release frequencies — is an open methodological question. It is deferred deliberately, with an explicit caution: any such integration must not reintroduce attack-probability estimation through the back door, because doing so would forfeit the deterministic discipline that gives the method its regulatory acceptability (Section 4.2).

Amended at version 0.2. The independence-class count of Section 3.2 is a quantification that is not a likelihood. It is a quantity with a stated derivation, produced from a declared dependency set and carrying no frequency claim, and it therefore meets this item's caution in the one form the caution permits: it measures the redundancy a configuration has against an adversary who selects correlations without estimating how probable any adversary action is. It does not answer this item's question. How such a quantity should sit beside core-damage-frequency and large-early-release-frequency results in a risk-informed integration remains open, and the caution stands against any answer that would attach a frequency to the count.

§7.4–§7.5 — Nuclear-specific material, moved at v0.3 to DBA-NUC-DiD §7.2–§7.3.

7.6 Reporting Convention for the Bounded Class Count

Section 3.2 fixes the form in which the independence-class count is reported, a bounded range with its deciding judgments named, and does not yet fix the convention for producing it. Three things are open: how many judged entries are varied when a declaration carries many, and whether they are varied jointly or one at a time; how confidence in each judgment is stated; and how the reviewer Section 3.2 requires records that a judgment was tested rather than accepted. The question arose in the application of the method in *The Asset and the Target*, the DBA-EN companion paper that recommends and applies the bounded-range form. Until a convention is set, a declaration reports both bounds, names every judgment that moves the count between them, and records who reviewed it.

7.7 Implementation Diversity as Independent Protection

Section 3.4 does not yet rule on whether genuinely diverse implementations of a credited function, such as distinct control implementations, count as independent protection of a governing cutset. Its independence test supplies a reading: diversity counts where the implementations share no codebase, vendor, toolchain, or update path, because no single action

set then defeats both, and it does not count where they share any of these, because the shared element is then substrate the ledger must record. Under that reading diversity is not an added redundant barrier, which Section 3.4 rejects, but a change to the Section 3.2 ledger in which one shared element is replaced by several. The reading has been applied in *The Asset and the Target* and has not been adopted here. A ruling is needed before a facility-class author first credits two such implementations.

7.8 Sufficiency of the Four Dependency Classes Across Sectors

The four dependency classes of §3.2 — physical, electrical, digital and organizational — were written against nuclear facilities. Whether they suffice in every sector, or whether some sectors need a further class, is open. Two candidates are visible. One is a supply class, for the upstream materials, spares and replacement equipment on which recovery depends, which the class-level evaluation of §3.4 already reaches through recovery. The other is a logical class, for the data, identity and transaction substrate of the financial and information technology sectors. Until a ruling is made, an author records such a dependency in the class it most nearly fits and names the fit as a judgment, so that it enters the bounded range of §3.2 rather than escaping it.

Revision History

Version	Date	Author	Description
0.1	July 2026	T. Roxey	Initial working draft. Establishes Defense-in-Depth Under Military Action as a cross-cutting methodological instrument of the DBA-MA series, peer to DBA-MA-ST and referenced by the facility-class, use-case, and national-child documents rather than re-derived in each. Develops the barrier-independence ledger (four dependency classes), the adversary-selected cutset analysis (minimal substrate-defeat sets screened for feasibility against the design-basis threat), and the independent-protection adequacy criterion, yielding a deterministic, bounding DiD-adequacy statement at the safety-security interface. Positions the method as an extension of probabilistic-risk-assessment common-cause failure analysis rather than a replacement, and as expressible under IAEA SSR-2/1 and INSAG-10. Worked against the DBA-MA-SMR-FC1 multi-module campus bounding scenario, reproducing that document's declared residual from the barrier-correlation side; Zaporizhzhia used to validate, not to define. Working document tag DBA-MA-DiD pending confirmation.
0.2	22 September 2026	T. Roxey	Receives three elements accepted, under CR-OT §5, from the Project 8 harvest against the correlated-barrier cutset treatment. Source instrument: *The Asset and the Target* (DBA-EN companion paper), by way of Project 8's note of 14 September 2026, *The Cutset Doctrine Diagnoses and Has No Disposition Step*. Determined in Project 6's note to Project 8 of 14 September 2026, *The Parent Already Carries a Disposition Step at DBA-MA-DiD Section 3.4*, §6, whose wording is used as issued. By element: (1) the independence-class count by transitive closure over the declared dependency set, from Project 8's closure definition, enters §3.2 as the output of Move 1; (2) the declarant discipline, from Project 8's note §11, which named for the count an exposure the Move 1 ledger already had and did not name, is folded into the same §3.2 passage; (3) the class-level consequence evaluation, from Project 8's note §4, with its recovery sentence carried from that note's §6, enters §3.4 before the residual sentence and ends in this instrument's two-outcome disposition. The four-way disposition step offered in the same harvest is declined because §3.4 already carries one. One change to the determined text: the second §3.2 paragraph, reporting the count as a bounded range with its deciding judgments named, is the author's ruling of 22 September 2026 on the fourth question in Project 8's note of 16 September 2026, and Project 8 is told. Amends §7.1 (the declared dependency set now has an output and a discipline; template registration remains open) and §7.3 (the count is a quantification that is not a likelihood and meets the caution; the integration question remains open). Adds §7.6, the reporting convention for the bounded count, and §7.7, implementation diversity as independent protection, both open. CR-OT §5

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			conditions one and two were performed in Project 8's instrument, and condition three by Project 8's dependents reading of 15 September 2026, which found one dependent family; this entry performs condition four.
0.3	7 November 2026	T. Roxey	Extraction under CR-OT (*Ownership Transfer*), determined by Project 0 in *Defense in Depth Is Sector-Neutral, and Reference Events May Offer It* (7 November 2026, version 0.2), on the author's rulings of the same date. The method becomes sector-neutral shared machinery of the DBA-MA extension and keeps its designation and title for the same referent; the nuclear application leaves to DBA-NUC-DiD v0.1, issued the same day, which this instrument now cites for it. Moved out, each leaving a one-line note at its number: §4.3 to DBA-NUC-DiD §3; §4.4 to DBA-NUC-DiD §4; the introduction to Section 5 and §5.1 to §5.3, with the governing-cutset table, to DBA-NUC-DiD Section 5 and §5.1 to §5.3, Project 6's allocation of the worked illustration, which the determination left to it, being that the illustration goes with the nuclear application; §5.4 to DBA-NUC-DiD §5.4; §6.1 to §6.3 to DBA-NUC-DiD §6.1 to §6.3; §7.4 and §7.5 to DBA-NUC-DiD §7.2 and §7.3; and the nuclear sequencing sentence of §7.2 to DBA-NUC-DiD §7.1. Added at new numbers: §3.6 (robust, diverse and independent, and the distinction from resilient), §4.5 (a sector's adversarial design basis and consequence analysis), §5.5 (worked examples in the sectors), §6.4 (application across the sectors, including the reference-event countermeasure) and §7.8 (sufficiency of the four dependency classes). Restated in sector-neutral terms without change of method: the cover, which drops the nuclear subtitle and series line; §1.1 to §1.4 and the inheritance table; §2.1, §2.2.2, §2.2.3 and §2.3; §3.1, §3.3 and the first paragraph of §3.4, which also gains a closing sentence pointing to §3.6; §3.5; §4.1; and the statements of §7.2 and §7.3. A note on this version is added before Section 1, and Section 5 is retitled Worked Illustrations. Travel with the method unchanged: §7.6 and §7.7. Provenance kept: the passages of §3.2 and §3.4 received from *The Asset and the Target* on 22 September 2026 under the transfer recorded at version 0.2 stand verbatim, and the version 0.2 entry's account of them is unchanged, so that this instrument, the receiver of that transfer, keeps the record of how it received while becoming the source of this one. The amendment records within §7.1 and §7.3 are unchanged. Item B4 of the NRC submittal package, which holds version 0.1, is not disturbed. The title of DBA-MA-PPA, which version 0.2 gave at §1.4 and §7.4 as a physical-protection assessment methodology, is given correctly in DBA-NUC-DiD as *Physical Proximity Attack*.
0.4	8 November 2026	T. Roxey	Consolidation of the vacated-number notes under the first condition of the second determination as amended in Defense in Depth Is Sector-Neutral, and Reference Events May Offer It (version 0.4, 8

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			November 2026), on the author's direction given in Project 0's note of 8 November 2026 (d). The twelve one-line stubs of version 0.3, with the headings of the vacated subsections under which they sat, are replaced by four range notes: §4.3–§4.4 in Section 4 after §4.2; §5.1–§5.4, which also takes the place of the Section 5 introduction stub, at the head of Section 5 before §5.5; §6.1–§6.3 in Section 6 before §6.4; and §7.4–§7.5 in Section 7 between §7.3 and §7.6. Each note names every number in its run and says moved at v0.3, because that is when the text moved. No section is renumbered. No text changes in any section that states the method, the Note on This Version keeps its full section-by-section map, and the partial move out of §7.2 stays as the sentence already in that section.